

Playing Tougher for One's Size: Positional Physicality in the NHL

Rento Saijo '28

Statistics and Data Science, Computer Science
Data, Information, and Society Pathway
Funded Sophomore Research Program

Jeff Moher, Connecticut College; Sian Lee, Michigan State University; Yeongjun Seo, U. of Tennessee



CONNECTICUT
COLLEGE



What does CSAx mean?

CSAx measures physical presence relative to a player's listed height and weight.

DIRECT CONTACT

Hits given or taken, blocks, fights, contact penalties

PUCK + SPACE

Net-front plays and forechecking; defensive recoveries and entry denials

HIGHER CSAx More physical presence for one's size

LOWER CSAx Less physical presence for one's size

2024-25 Top 3 CSAx

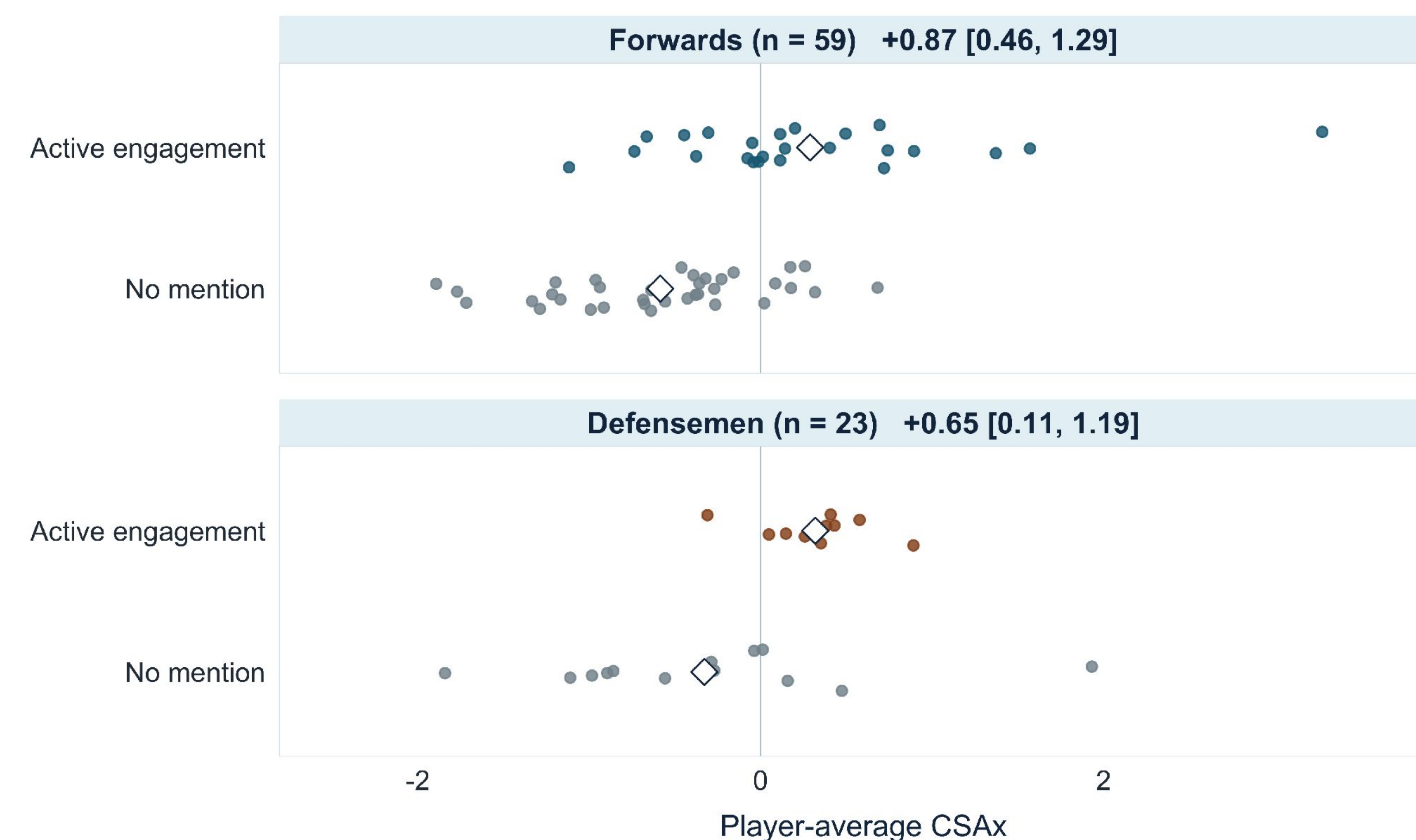
FORWARDS			DEFENSEMEN		
#	Player	CSAx	#	Player	CSAx
1	Sam Carrick	4.69	1	Connor Clifton	2.95
2	Mathieu Olivier	3.88	2	Radko Gudas	2.93
3	Zachary L'Heureux	3.28	3	Nick Seeler	2.82

Does it match what scouts see?

Scouting Report Example: Will Cuyllé

"Powerful skater with good speed. ... Heavy on the body and on pursuit of pucks. Very strong puck protection and wall play. Excellent net-front presence."

2020 Draft Guide, NHL Central Scouting



Each dot is a player; diamonds mark group means. Brackets show 95% intervals. Active-engagement descriptions align with higher CSAx in both positions.

Where does it matter?

Special-Teams Ice Time

Share of total ice time; CSAx +1 vs. -1

	POWER PLAY	PENALTY KILL
Forwards	-1.2 points	+1.4 points
Defensemen	-0.9 points	+1.3 points

Playoff Hitting among Forwards

First 4 playoff games vs. regular-season baseline

Lower CSAx	+136%
Higher CSAx	+84%

Hits rise in both forward profiles, with a smaller proportional rise for higher-CSAx players; the defenseman contrast is uncertain.

Next-Season Roster Continuation

Chance of reaching 300 NHL minutes; CSAx +1 vs. -1

Forwards	+2.9 percentage points
Defensemen	+4.2 percentage points

Among returners, higher CSAx shows no clear increase in minutes per game, so it accompanies staying power but not necessarily larger role.



Summer 2026 NHL Intern Cohort

Behind This Study

During my NHL Stats R&D internship, I discussed this research with supervisors and colleagues whose hockey and analytics perspectives helped sharpen our questions. I thank them for their guidance as this work took shape. We plan to present at Carnegie Mellon, submit to MIT Sloan, and develop a journal manuscript.

Data Sources

We combine NHL play-by-play and official ice-time records with Corey Sznajder's All Three Zones tracking data. The NHL records describe contact and playing time, while the tracking data adds forechecking and defensive-zone actions. NHL Central Scouting profiles provide independent descriptions of physical engagement for comparison with CSAx.



Full 2024-25
Rankings and
Technical Details